

Lab 1: Packet analysis at application layer using Wireshark
SCSR1213 Network Communications
Universiti Teknologi Malaysia

Objective:

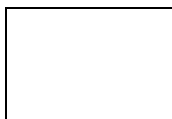
1. Understanding of network protocols by observing the sequence of messages exchanged between two protocol entities, delving down into the details of protocol operation, and causing protocols to perform certain actions and then observing these actions and their consequences.
2. To introduce student with Wireshark software tool for packet analyzer.
3. To analyze protocol used in application layer such as http and dns.

Reference material: Computer Networking: A Top-Down Approach, 7th ed., J.F. Kurose and K.W. Ross.

Name : FARRA NURZAHIN BINTI ZAHARIL ANUAR

Metric No : A23CS0079

Section : 02



Mark

PART A: Wireshark Getting Started

1.0 Introduction

The basic tool for observing the messages exchanged between executing protocol entities is called a **packet sniffer**. As the name suggests, a packet sniffer captures (“sniffs”) messages being sent/received from/by your computer; it will also typically store and/or display the contents of the various protocol fields in these captured messages. A packet sniffer itself is passive. It observes messages being sent and received by applications and protocols running on your computer, but never sends packets itself. Similarly, received packets are never explicitly addressed to the packet sniffer. Instead, a packet sniffer receives a *copy* of packets that are sent/received from/by application and protocols executing on your machine.

Figure A.1 shows the structure of a packet sniffer. At the right of Figure 1 are the protocols (in this case, Internet protocols) and applications (such as a web browser or ftp client) that normally run on your computer. The packet sniffer, shown within the dashed rectangle in Figure A.1 is an addition to the usual software in your computer, and consists of two parts. The **packet capture library** receives a copy of every link-layer frame that is sent from or received by your computer. In Figure A.1, the assumed physical media is an Ethernet, and so all upper-layer protocols are eventually encapsulated within an Ethernet frame. Capturing all link-layer frames thus gives you all messages sent/received from/by all protocols and applications executing in your computer.

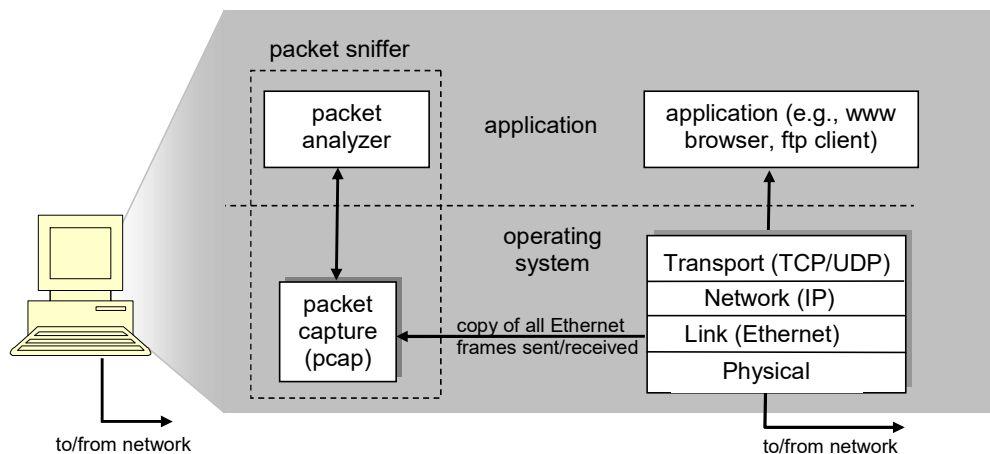


Figure A.1: Packet sniffer structure

The second component of a packet sniffer is the **packet analyzer**, which displays the contents of all fields within a protocol message. In order to do so, the packet analyzer must “understand” the structure of all messages exchanged by protocols. The packet analyzer understands the format of Ethernet frames, and so can identify the IP datagram within an Ethernet frame. It also understands the IP datagram format, so that it can extract the TCP segment within the IP datagram. Finally, it understands the TCP segment structure, so it can extract the HTTP message contained in the TCP segment. Finally, it understands the HTTP protocol and so, for example, knows that the first bytes of an HTTP message will contain the string “GET,” “POST,” or “HEAD”.

2.0 Getting Wireshark Ready

- Download and install the Wireshark software
- Run Wireshark. Wireshark startup screen shown in Figure A.2.

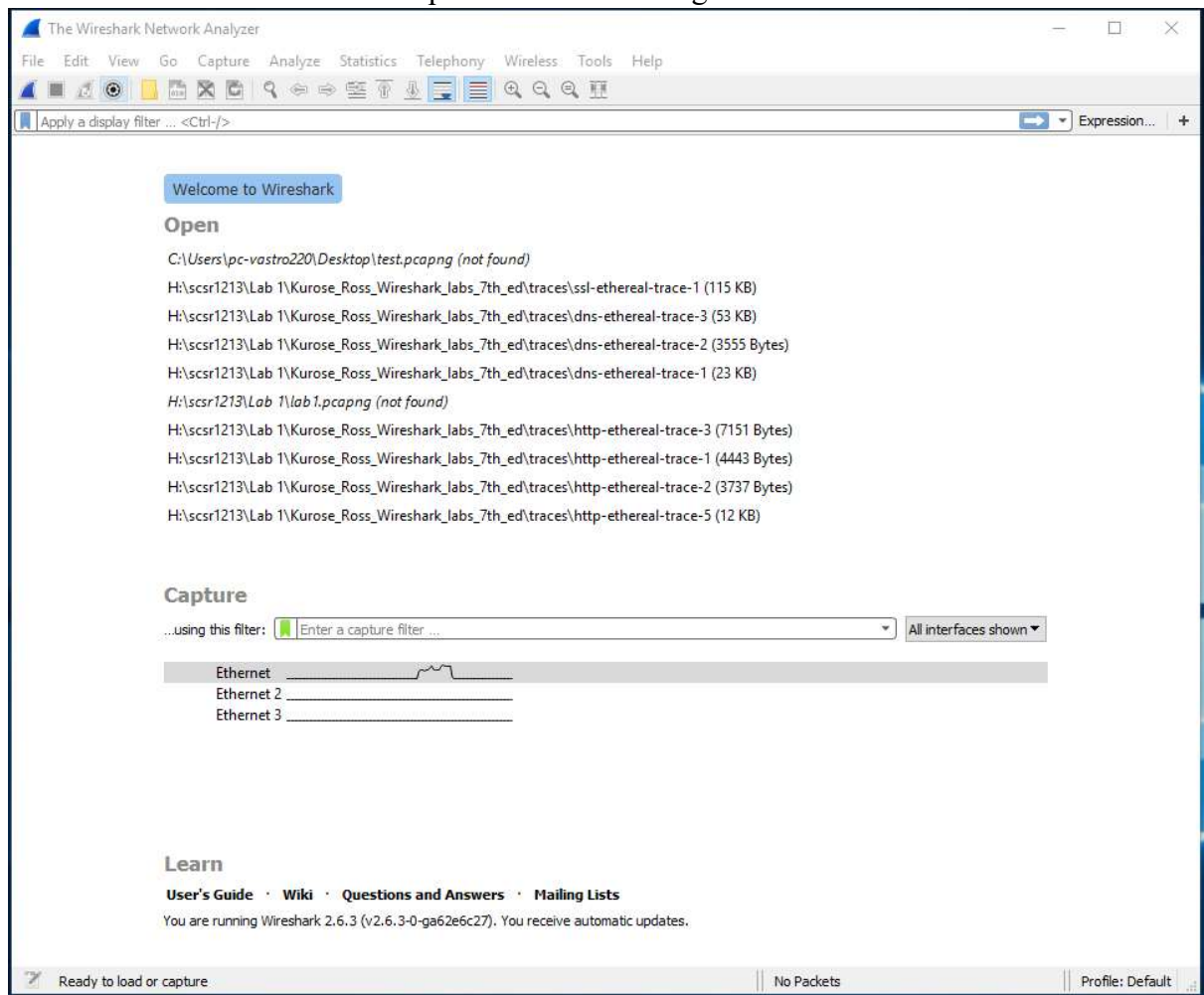


Figure A.2: Initial Wireshark startup screen

- The Wireshark interface has five major components as shown in Figure A.3.

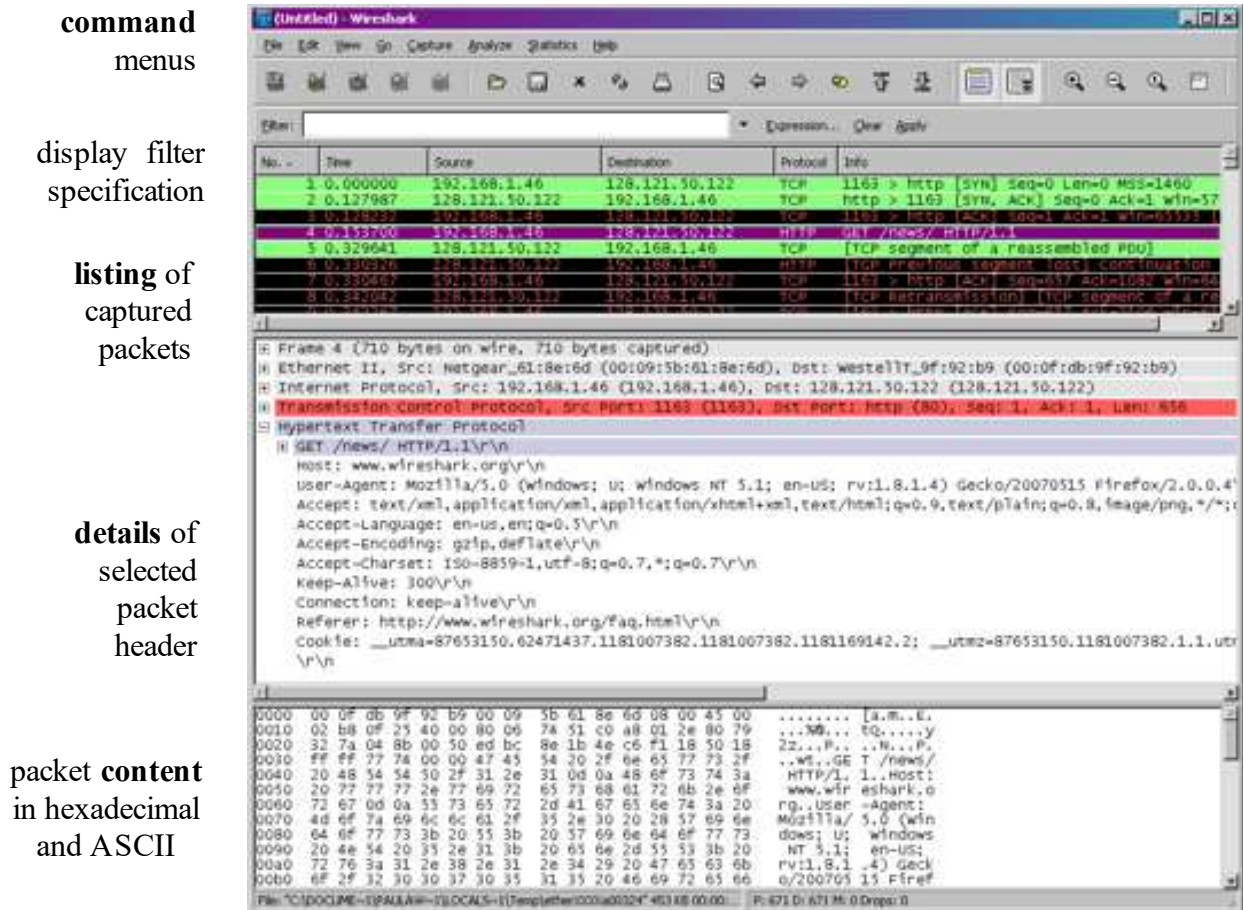


Figure A.3: Wireshark Graphical User Interface, during packet capture and

- The **command menus** are standard pulldown menus located at the top of the window.
- The **packet display filter field**, into which a protocol name or other information can be entered in order to filter the information displayed in the packet-listing window.
- The **packet-listing window** displays a one-line summary for each packet captured, including the packet number, the time at which the packet was captured, the packet's source and destination addresses, the protocol type, and protocol-specific information contained in the packet.
- The **packet-header details window** provides details about the packet selected (highlighted) in the packet-listing window. These details include information about the Ethernet frame and IP datagram that contains this packet. The amount of Ethernet and IP-layer detail displayed can be expanded or minimized by clicking on the plus minus boxes to the left of the Ethernet frame or IP datagram line in the packet details window. If the packet has been carried over TCP or UDP, TCP or UDP details will also be displayed, which can similarly be expanded or minimized. Finally, details about the highest-level protocol that sent or received this packet are also provided.
- The **packet-contents window** displays the entire contents of the captured frame, in both ASCII and hexadecimal format.

3.0 Test Run Wireshark

- Start up the Wireshark software.
- To begin packet capture, select the Capture pull down menu and pick Options menu. Select appropriate interfaces on your compute and click Start button to begin packet capture. Refer to Figure A.4

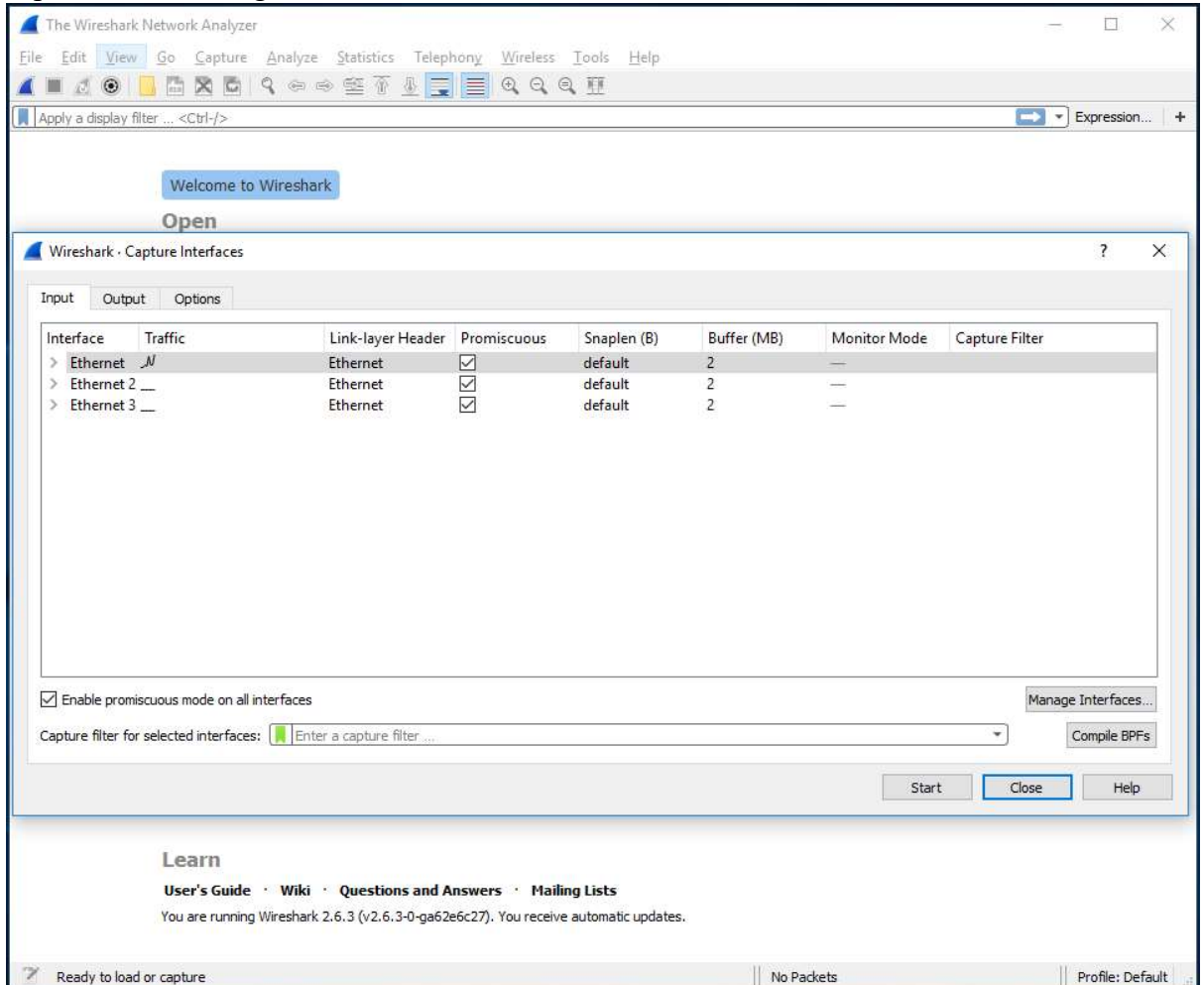


Figure A.4: Capture and Options Menu

- Once you begin packet capture, result will be shown as in Figure A.5.

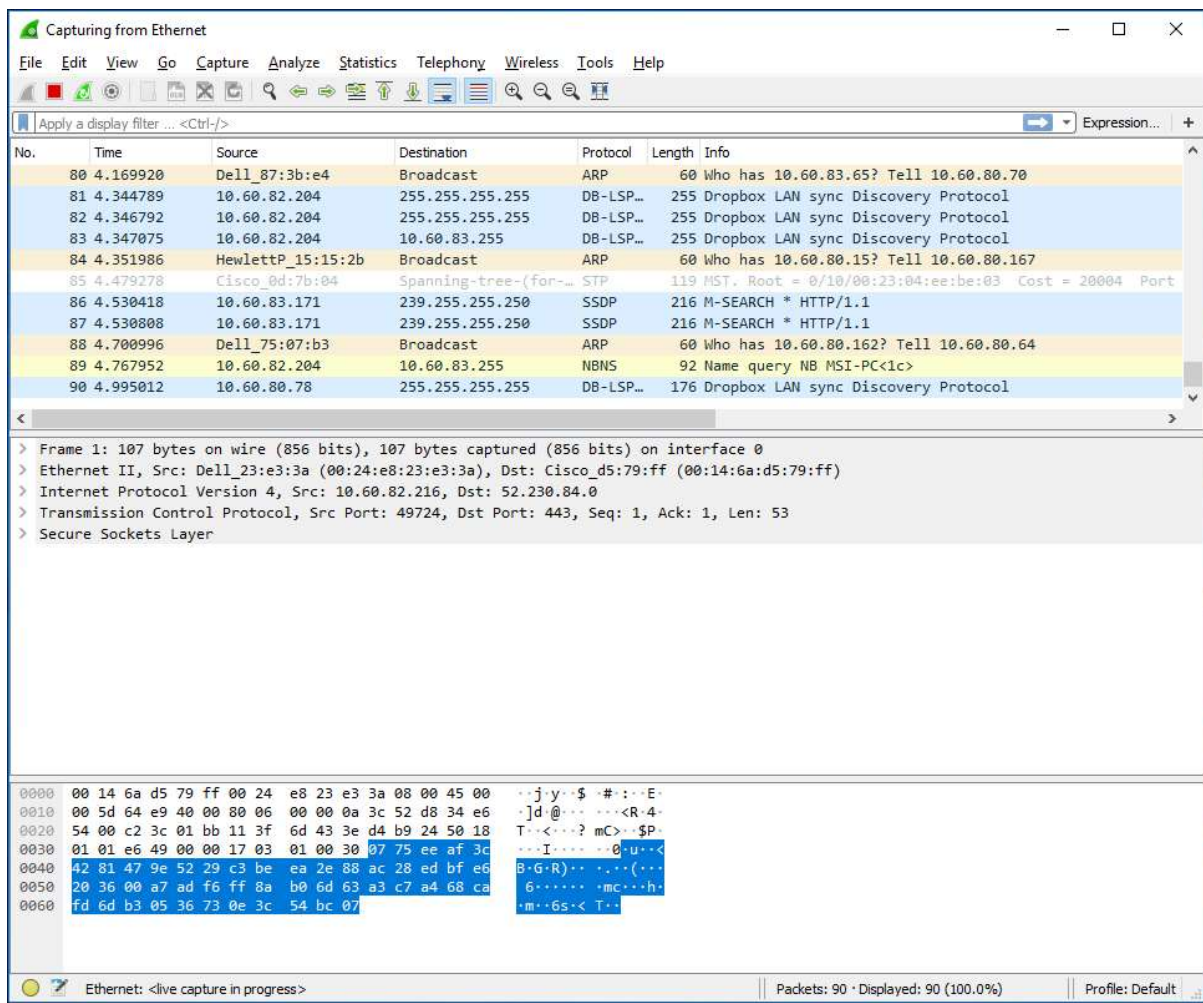


Figure A.5: Wireshark packet capture result

- By selecting Capture pulldown menu and selecting Stop, you can stop packet capture.

- Type “arp” in packet display filter field and press Enter key. This will cause only ARP message to be displayed in the packet-listing window as shown in Figure A.6.

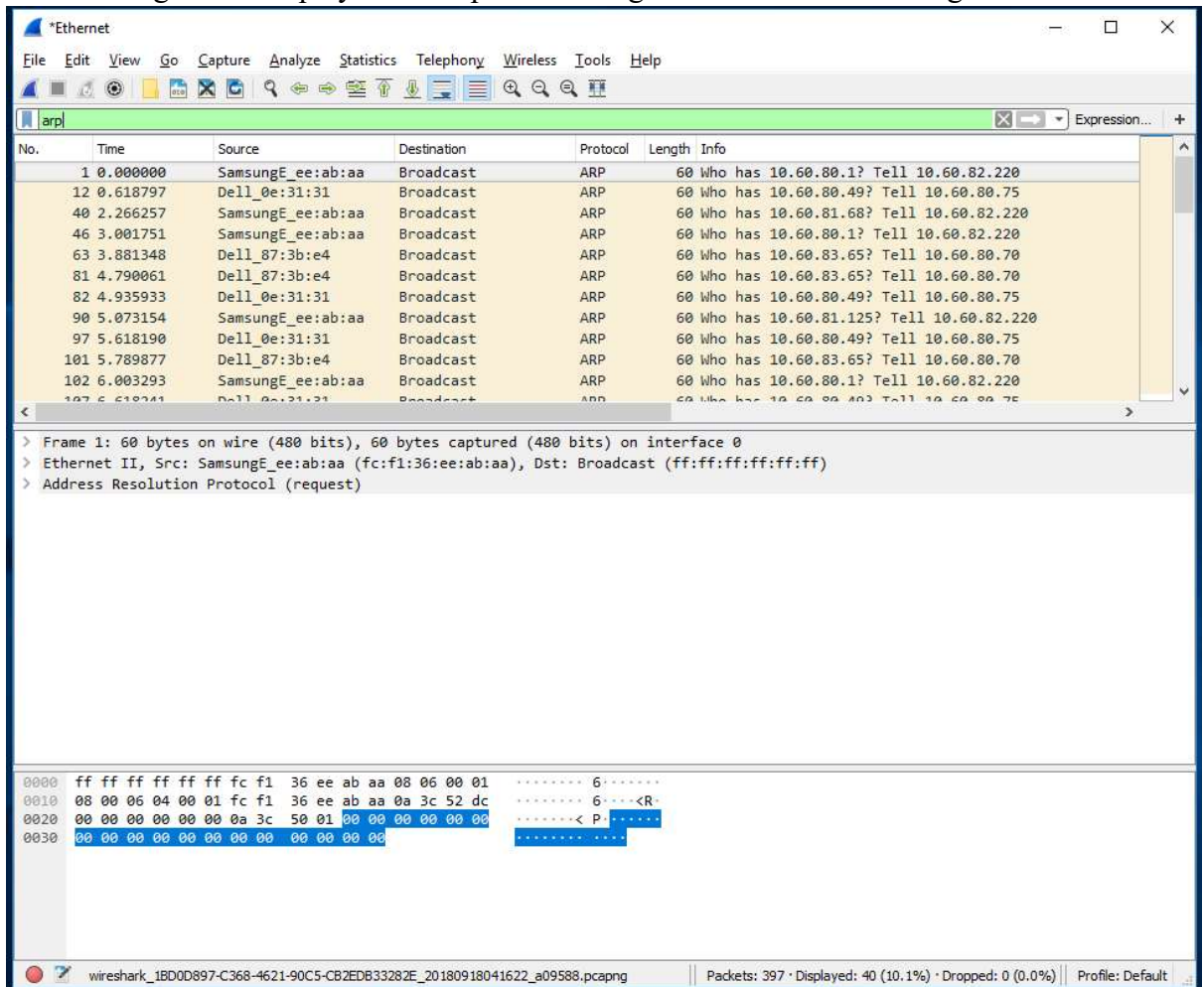


Figure A.6: ARP packet capture

- To save the trace result, use File pulldown menu and select Save function as shown in Figure A.7.

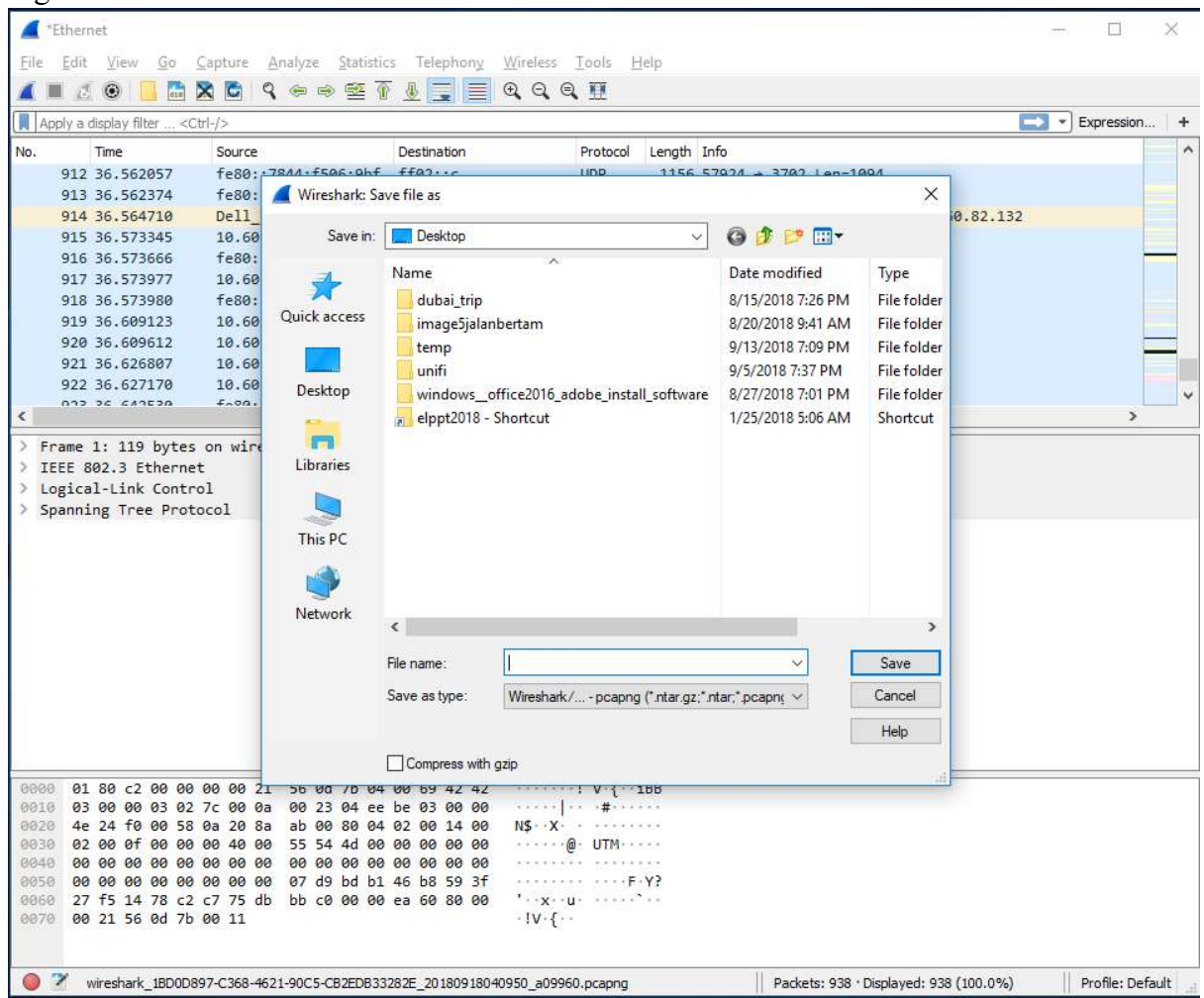


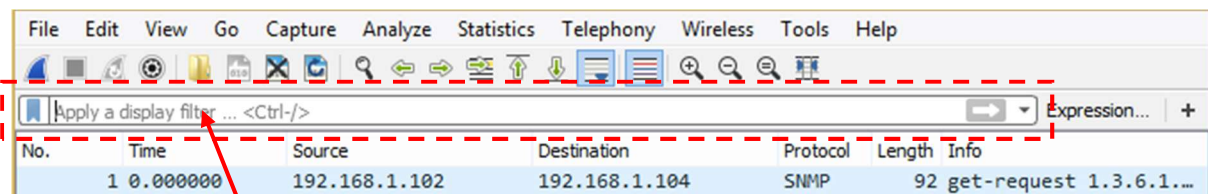
Figure A.7: Save Wireshark trace result

PART B: HTTP Trace

In this part, we'll explore several aspects of the HTTP protocol: the basic GET/response interaction, HTTP message formats and retrieving HTML files with embedded objects. Before beginning these labs, you might want to review Section 2.2 of the textbook.

B.1 The Basic HTTP GET/response interaction

- Open packet trace file **lab1-http-B01.pcapng**.
- Enter “**http**” (just the letters, not the quotation marks) in the **packet display filter field**, so that only captured HTTP messages will be displayed later in the packet-listing window. Refer to figure below:



packet display filter

- By looking at the information in the HTTP GET and response messages, answer the following questions:
 1. What version of HTTP is the server running?
The server is running HTTP/1.1
 2. What is the IP address of the client computer?
Client computer IP address is 192.168.1.102
 3. What is the IP address of the gaia.cs.umass.edu server?
gaia.cs.umass.edu serve IP address is 128.119.245.12
 4. How many bytes of content are being returned to client browser?
The packet with the 200 OK response has a length of 439 bytes
 5. What is the status code returned from the server to client browser?
The response contains 200 OK and 404 Not Found

B.2 The HTTP CONDITIONAL GET/response interaction

- Open packet trace file **lab1-http-B02.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. Inspect the contents of the first HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE” line in the HTTP GET?

No, the first HTTP GET request does not contain an “IF-MODIFIED-SINCE” header.

2. Inspect the contents of the server response after the first GET request from client. Did the server explicitly return the contents of the file? How can you tell?

Yes, the server returned the contents of the file. This can be confirmed by the HTTP/1.1 200 OK response and the 739 bytes of content in the response.

3. Now inspect the contents of the second HTTP GET request from your browser to the server. Do you see an “IF-MODIFIED-SINCE:” line in the HTTP GET? If so, what information follows the “IF-MODIFIED-SINCE:” header?

No, it says 304 Not modified. However, this status code indicates that client likely sent an “IF-MODIFIED-SINCE” header in the request.

4. What is the HTTP status code and phrase returned from the server in response to this second HTTP GET? Did the server explicitly return the contents of the file? Explain.

The status code and phrase returned from the server is HTTP/1.1 304 Not Modified. This means the server did not send the file contents again, indicating that the file has not changed since the last transmission.

B.3 HTML Documents with Embedded Objects

- Open packet trace file **lab1-http-B03.pcapng**.
- By looking at the information in the HTTP GET and response messages, answer the following questions:

1. How many HTTP GET request messages did client browser send?

Three

2. To which Internet addresses were these GET requests sent?

The GET requests were sent to the following destination IP addresses:

- 128.119.245.12
- 165.193.123.218
- 134.241.6.82

3. any bytes of content are being returned to client browser for the **pearson-logo-footer.gif** image file?

Yes, 625 bytes

4. How many bytes of content are being returned to client browser for the **cover.jpg** image file?

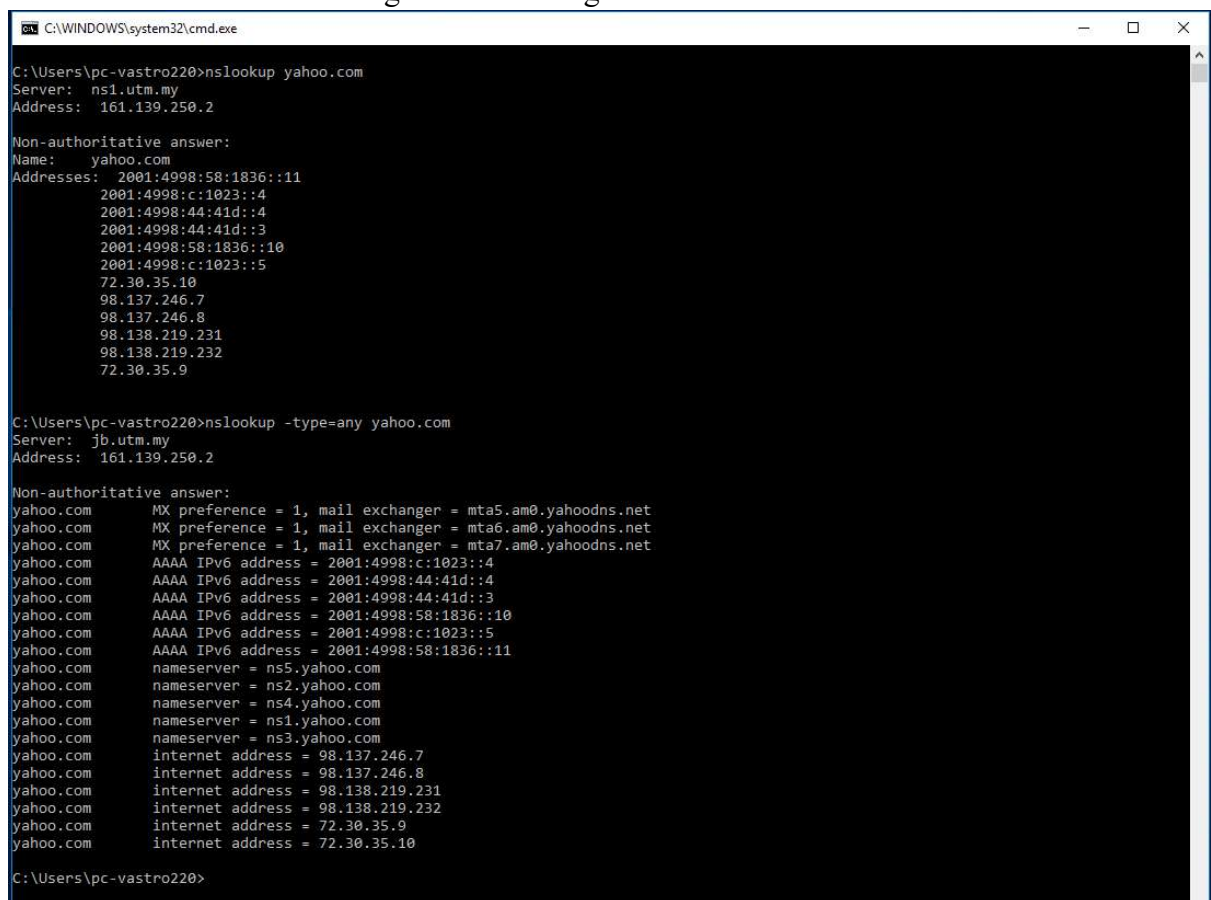
609 bytes.

PART C: DNS Trace

1.0 nslookup

nslookup tool allows the host running the tool to query any specified DNS server for a DNS record. The queried DNS server can be a root DNS server, a top-level-domain DNS server, an authoritative DNS server, or an intermediate DNS server. To accomplish this task, nslookup sends a DNS query to the specified DNS server, receives a DNS reply from that same DNS server, and displays the result.

- To run it in Windows, open the Command Prompt (cmd) and run nslookup on the command line as shown in Figure C.1 and Figure C.2



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup yahoo.com
Server: ns1.utm.my
Address: 161.139.250.2

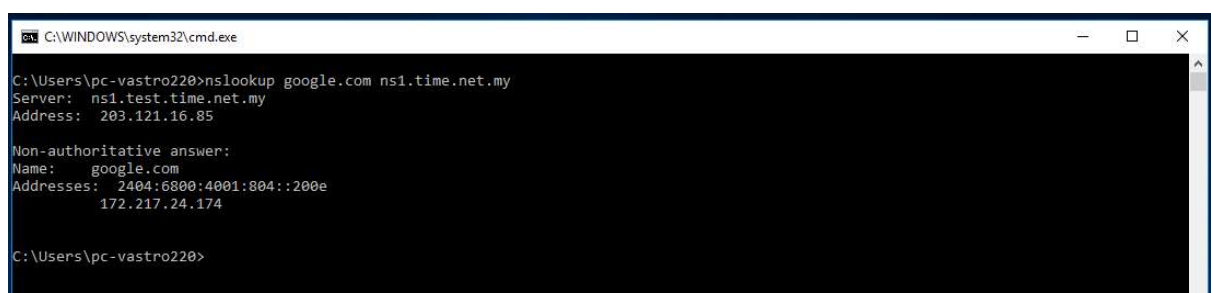
Non-authoritative answer:
Name: yahoo.com
Addresses: 2001:4998:58:1836::11
           2001:4998:c:1023::4
           2001:4998:44:41d::4
           2001:4998:44:41d::3
           2001:4998:58:1836::10
           2001:4998:c:1023::5
           72.30.35.10
           98.137.246.7
           98.137.246.8
           98.138.219.231
           98.138.219.232
           72.30.35.9

C:\Users\pc-vastro220>nslookup -type=any yahoo.com
Server: jlb.utm.my
Address: 161.139.250.2

Non-authoritative answer:
yahoo.com      MX preference = 1, mail exchanger = mta5.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta6.am0.yahoodns.net
yahoo.com      MX preference = 1, mail exchanger = mta7.am0.yahoodns.net
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::4
yahoo.com      AAAA IPv6 address = 2001:4998:44:41d::3
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::10
yahoo.com      AAAA IPv6 address = 2001:4998:c:1023::5
yahoo.com      AAAA IPv6 address = 2001:4998:58:1836::11
yahoo.com      nameserver = ns5.yahoo.com
yahoo.com      nameserver = ns2.yahoo.com
yahoo.com      nameserver = ns4.yahoo.com
yahoo.com      nameserver = ns1.yahoo.com
yahoo.com      nameserver = ns3.yahoo.com
yahoo.com      internet address = 98.137.246.7
yahoo.com      internet address = 98.137.246.8
yahoo.com      internet address = 98.138.219.231
yahoo.com      internet address = 98.138.219.232
yahoo.com      internet address = 72.30.35.9
yahoo.com      internet address = 72.30.35.10

C:\Users\pc-vastro220>
```

Figure C.1: nslookup result



```
C:\WINDOWS\system32\cmd.exe

C:\Users\pc-vastro220>nslookup google.com ns1.time.net.my
Server: ns1.test.time.net.my
Address: 203.121.16.85

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4001:804::200e
           172.217.24.174

C:\Users\pc-vastro220>
```

Figure C.2: nslookup result

1. Run nslookup to obtain the IP address of a www.microsoft.com server. What is the IP address of that server? Add screenshot to your answer.

```
C:\Users\user>nslookup www.microsoft.com
Server: UnKnown
Address: fe80::805f:c5ff:fe39:2e64

Non-authoritative answer:
Name: e13678.dscb.akamaiedge.net
Addresses: 2404:160:100:ff92::356e
           2404:160:100:ff91::356e
           183.171.229.133
Aliases: www.microsoft.com
         www.microsoft.com-c-3.edgekey.net
         www.microsoft.com-c-3.edgekey.net.globalredir.akadns.net
```

2. Run nslookup to determine the non-authoritative DNS servers for domain microsoft.com. Add screenshot to your answer.

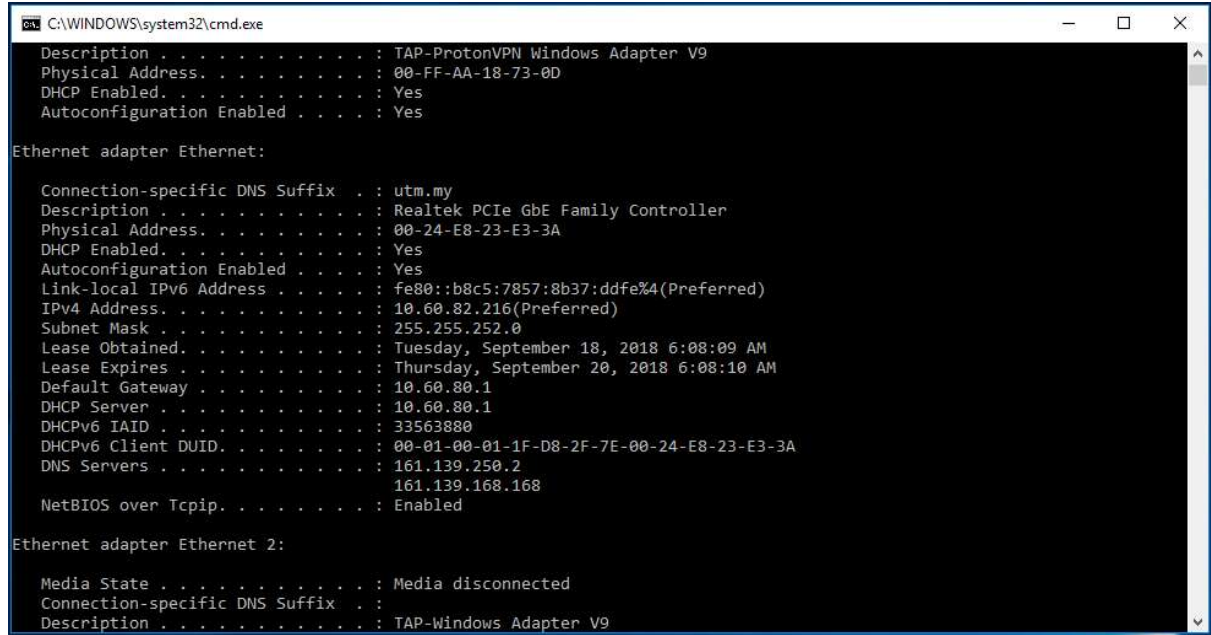
```
C:\Users\user>nslookup -type=ns microsoft.com
Server: UnKnown
Address: fe80::805f:c5ff:fe39:2e64

Non-authoritative answer:
microsoft.com nameserver = ns3-39.azure-dns.org
microsoft.com nameserver = ns4-39.azure-dns.info
microsoft.com nameserver = ns1-39.azure-dns.com
microsoft.com nameserver = ns2-39.azure-dns.net
```

2.0 ipconfig

ipconfig can be used to show your current TCP/IP information, including your address, DNS server addresses, adapter type and so on.

- Information about host, use the following command: ipconfig /all



```
C:\WINDOWS\system32\cmd.exe
Description . . . . . : TAP-ProtonVPN Windows Adapter V9
Physical Address. . . . . : 00-FF-AA-18-73-0D
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Ethernet adapter Ethernet:

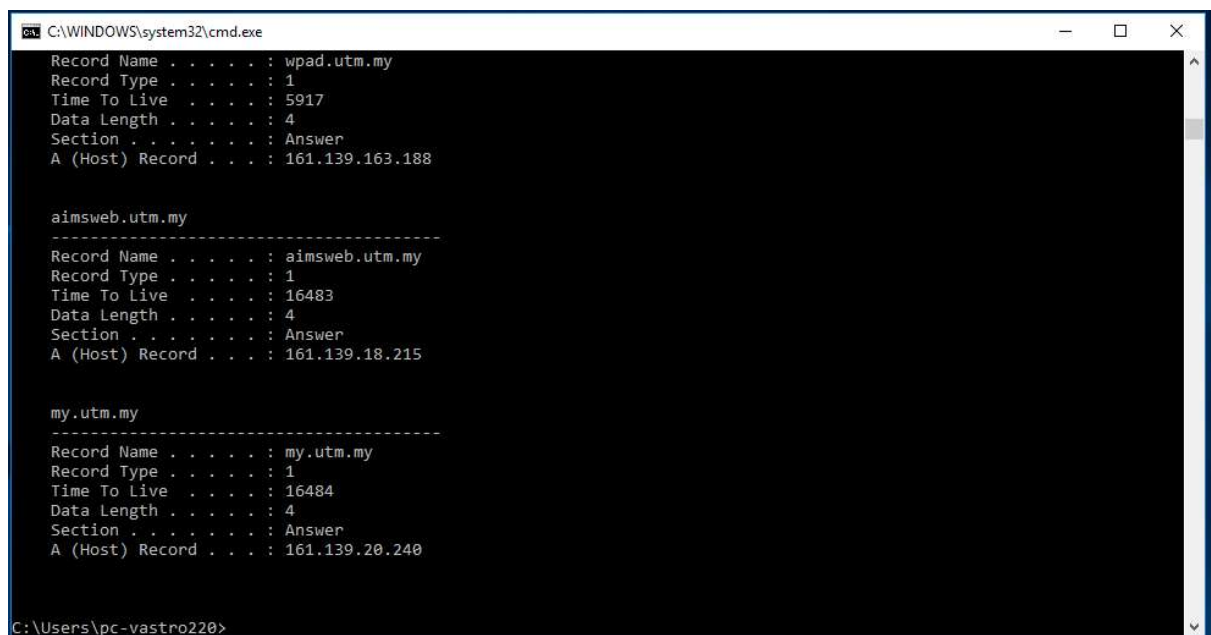
    Connection-specific DNS Suffix  . : utm.my
    Description . . . . . : Realtek PCIe GbE Family Controller
    Physical Address. . . . . : 00-24-E8-23-E3-3A
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::b8c5:7857:8b37:ddfe%4(Preferred)
    IPv4 Address. . . . . : 10.60.82.216(Preferred)
    Subnet Mask . . . . . : 255.255.252.0
    Lease Obtained. . . . . : Tuesday, September 18, 2018 6:08:09 AM
    Lease Expires . . . . . : Thursday, September 20, 2018 6:08:10 AM
    Default Gateway . . . . . : 10.60.80.1
    DHCP Server . . . . . : 10.60.80.1
    DHCPv6 IAID . . . . . : 33563880
    DHCPv6 Client DUID. . . . . : 00-01-00-01-1F-D8-2F-7E-00-24-E8-23-E3-3A
    DNS Servers . . . . . : 161.139.250.2
                           161.139.168.168
    NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Ethernet 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . :
    Description . . . . . : TAP-Windows Adapter V9
```

Figure C.3: ipconfig /all result

- ipconfig is also very useful for managing the DNS information stored in your host. Each entry shows the remaining Time to Live (TTL) in seconds.
Command: ipconfig /displaydns



```
C:\WINDOWS\system32\cmd.exe
Record Name . . . . . : wpad.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 5917
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.163.188

-----
aimsweb.utm.my
Record Name . . . . . : aimsweb.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16483
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.18.215

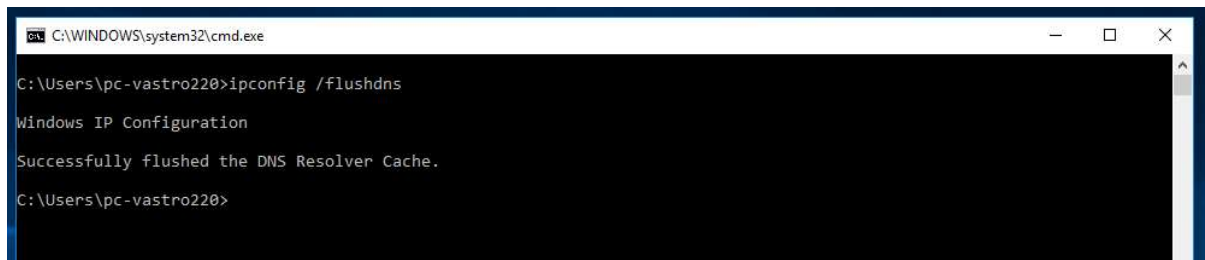
-----
my.utm.my
Record Name . . . . . : my.utm.my
Record Type . . . . . : 1
Time To Live . . . . . : 16484
Data Length . . . . . : 4
Section . . . . . : Answer
A (Host) Record . . . : 161.139.20.240

C:\Users\pc-vastro220>
```

Figure C.4: ipconfig /displaydns result

- Flushing the DNS cache clears all entries and reloads the entries from the hosts file.

Command: ipconfig /flushdns



```
C:\WINDOWS\system32\cmd.exe
C:\Users\pc-vastro220>ipconfig /flushdns

Windows IP Configuration

Successfully flushed the DNS Resolver Cache.

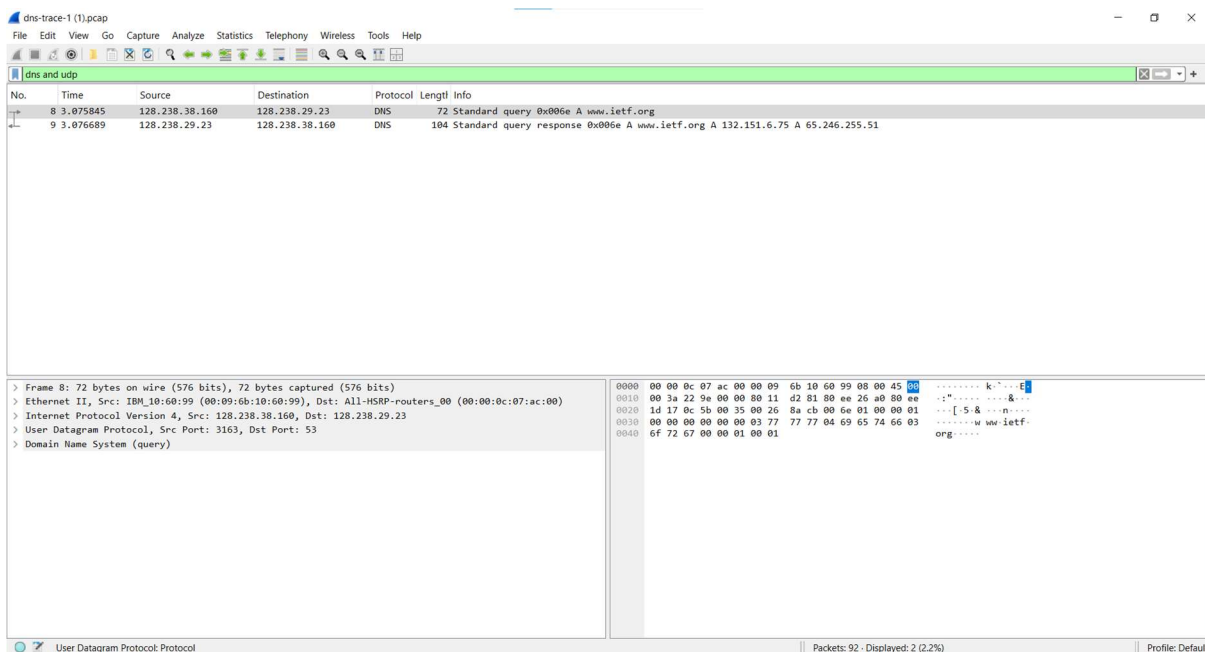
C:\Users\pc-vastro220>
```

Figure C.5: ipconfig /flushdns result

3.0 Tracing DNS with Wireshark

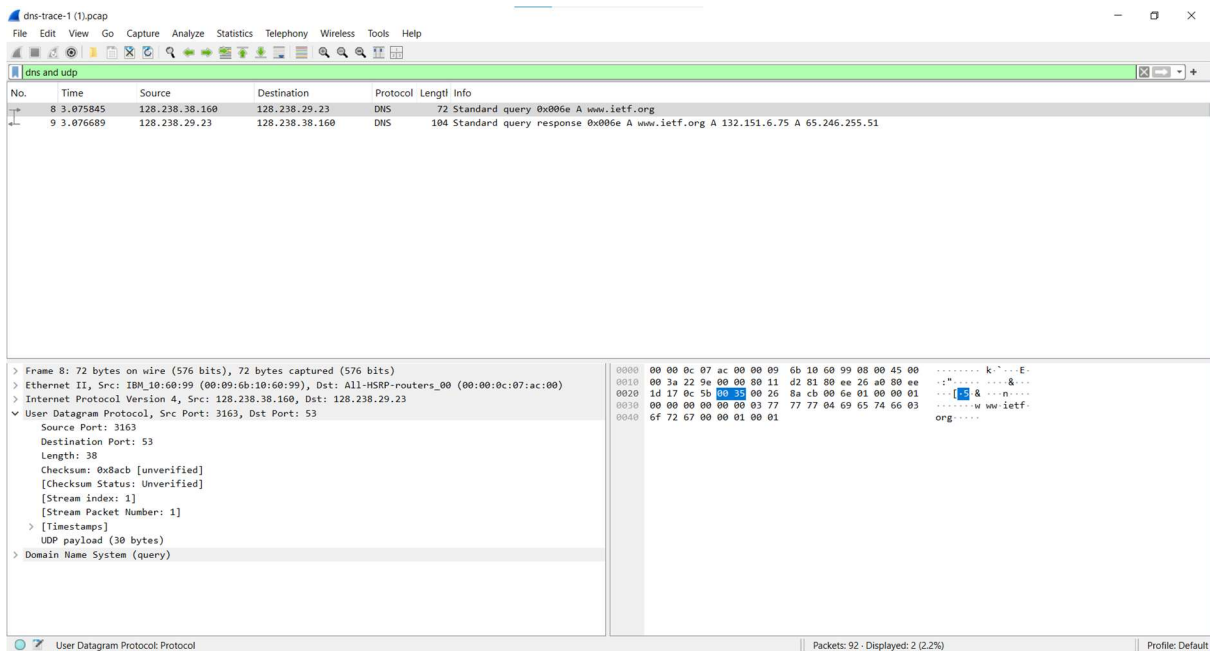
- Open packet trace file dns-trace-1. Answer the following questions.
 1. Locate the DNS query and response messages. Are then sent over UDP or TCP? Add screenshots in your answer.

UDP



2. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

Destination port for DNS query message is 53. Source port of DNS response message is also 53.



3. To what IP address is the DNS query message sent? Add screenshots in your answer.

DNS query message sent to IP address: 128.238.29.23

```

Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.23
  0100 .... = Version: 4
  .... 0101 = Header Length: 20 bytes (5)
  > Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
    Total Length: 58
    Identification: 0x229e (8862)
  > 000. .... = Flags: 0x0
    ...0 0000 0000 0000 = Fragment Offset: 0
    Time to Live: 128
    Protocol: UDP (17)
    Header Checksum: 0xd281 [validation disabled]
    [Header checksum status: Unverified]
    Source Address: 128.238.38.160
    Destination Address: 128.238.29.23
    [Stream index: 1]

```

4. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

Type = “A”.

The query message does not contain any “answers”.

```

  ▾ Domain Name System (query)
    Transaction ID: 0x006e
    > Flags: 0x0100 Standard query
    Questions: 1
    Answer RRs: 0
    Authority RRs: 0
    Additional RRs: 0
    ▾ Queries
      ▾ www.ietf.org: type A, class IN
        Name: www.ietf.org
        [Name Length: 12]
        [Label Count: 3]
        Type: A (1) (Host Address)
        Class: IN (0x0001)
        \[Response In: 9\]

```

5. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

There are 2 “answers”.

```

  ▾ Domain Name System (response)
    Transaction ID: 0x006e
    > Flags: 0x8180 Standard query response, No error
    Questions: 1
    Answer RRs: 2
    Authority RRs: 0
    Additional RRs: 0
    > Queries
    ▾ Answers
      ▾ www.ietf.org: type A, class IN, addr 132.151.6.75
        Name: www.ietf.org
        Type: A (1) (Host Address)
        Class: IN (0x0001)
        Time to live: 1678 (27 minutes, 58 seconds)
        Data length: 4
        Address: 132.151.6.75
      ▾ www.ietf.org: type A, class IN, addr 65.246.255.51
        Name: www.ietf.org
        Type: A (1) (Host Address)
        Class: IN (0x0001)
        Time to live: 1678 (27 minutes, 58 seconds)
        Data length: 4
        Address: 65.246.255.51
        \[Request In: 8\]
        [Time: 0.000844000 seconds]

```

6. Consider the subsequent TCP SYN packet sent by your host. Does the destination IP address of the SYN packet correspond to any of the IP addresses provided in the DNS response message? Add screenshots in your answer.

Destination IP address of the SYN packet: 132.151.6.75

```

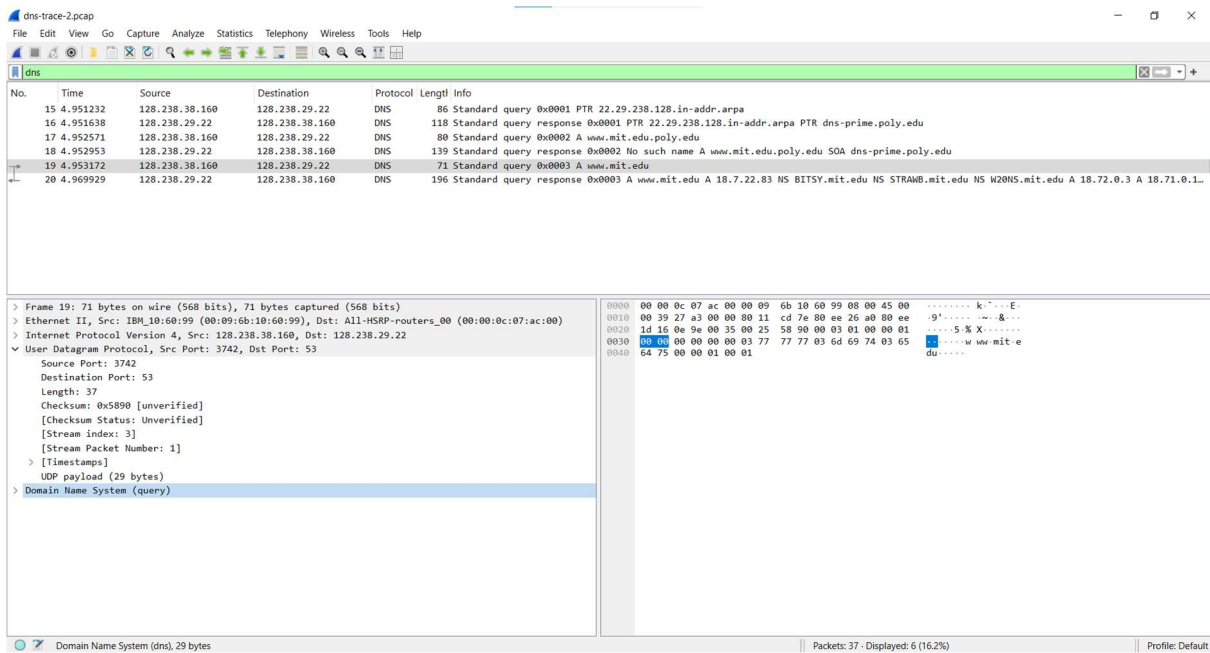
-
v Answers
  v www.ietf.org: type A, class IN, addr 132.151.6.75
    Name: www.ietf.org
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 1678 (27 minutes, 58 seconds)
    Data length: 4
    Address: 132.151.6.75
  v www.ietf.org: type A, class IN, addr 65.246.255.51
    Name: www.ietf.org
    Type: A (1) (Host Address)
    Class: IN (0x0001)
    Time to live: 1678 (27 minutes, 58 seconds)
    Data length: 4
    Address: 65.246.255.51
  [Request In: 8]
  [Time: 0.000844000 seconds]
```

7. This web page contains images. Before retrieving each image, does your host issue new DNS queries?

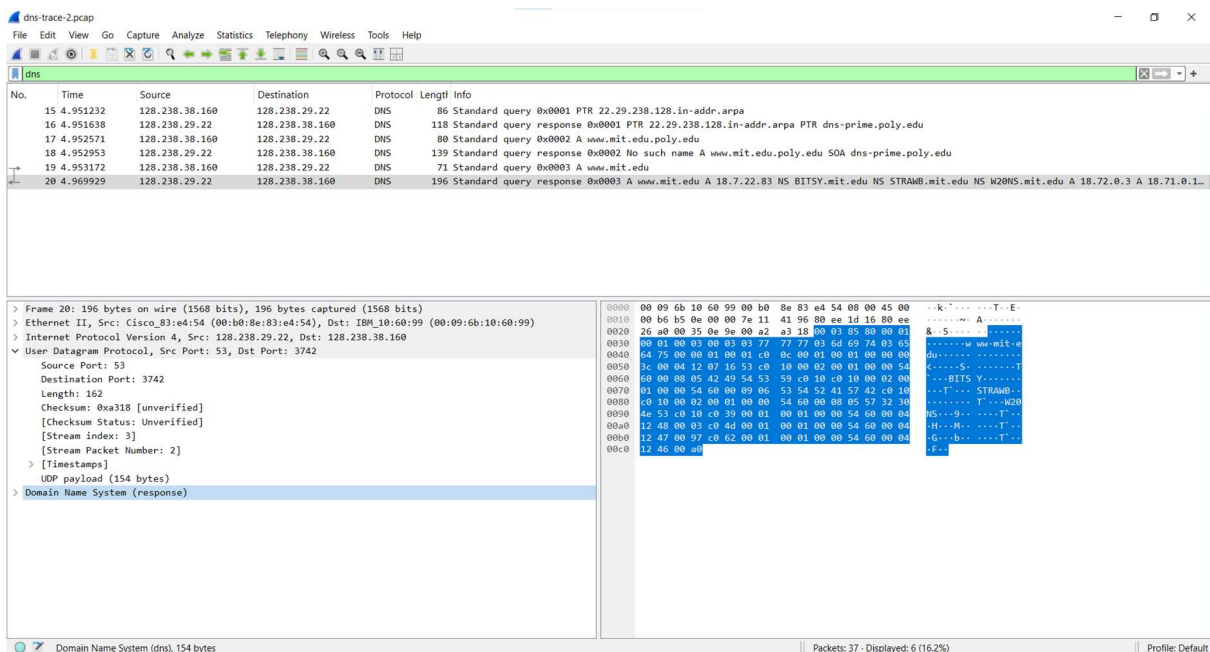
No

- Open packet trace file dns-trace-2 for nslookup.
 - We see from Wireshark that nslookup actually sent three DNS queries and received three DNS responses. For the purpose of this lab, ignore the first two sets of queries/responses, as they are specific to nslookup and are not normally generated by standard Internet applications. You should instead focus on the last query and response messages.
 - Answer the following questions.
8. What is the destination port for the DNS query message? What is the source port of DNS response message? Add screenshots in your answer.

Destination port DNS query message: 53



Source port of DNS response message: 53



- To what IP address is the DNS query message sent? Is this the IP address of your default local DNS server? Add screenshots in your answer.

IP address that the message was sent to is 128.238.29.22. However, the message is not sent to my default local DNS server.

My DNS server IP address: 172.20.10.1:

Wireless LAN adapter Wi-Fi:

```
Connection-specific DNS Suffix . :  
Description . . . . . : Intel(R) Dual Band Wireless-AC 8260  
Physical Address. . . . . : B4-69-21-0B-D1-32  
DHCP Enabled. . . . . : Yes  
Autoconfiguration Enabled . . . . : Yes  
IPv6 Address. . . . . : 2404:160:8058:979f:79ae:f402:5e59:495b(Preferred)  
Temporary IPv6 Address. . . . . : 2404:160:8058:979f:24e3:ddf4:9d44:2f16(Preferred)  
Link-local IPv6 Address . . . . . : fe80::40e4:a410:43ba:b5bb%11(Preferred)  
IPv4 Address. . . . . : 172.20.10.4(Preferred)  
Subnet Mask . . . . . : 255.255.255.240  
Lease Obtained. . . . . : Wednesday, 29 January, 2025 1:23:51 PM  
Lease Expires . . . . . : Thursday, 30 January, 2025 8:27:05 PM  
Default Gateway . . . . . : fe80::805f:c5ff:fe39:2e64%11  
                            172.20.10.1  
DHCP Server . . . . . : 172.20.10.1  
DHCPv6 IAID . . . . . : 269790149  
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-FF-83-38-E8-6A-64-C5-7E-03  
DNS Servers . . . . . : fe80::805f:c5ff:fe39:2e64%11  
                            172.20.10.1  
NetBIOS over Tcpi. . . . . : Enabled
```

DNS query message was sent to 128.238.29.22

The screenshot shows a Wireshark capture of a DNS query. The packet list on the left shows a DNS query from 128.238.38.160 to 128.238.29.22. The packet details on the right show the DNS query structure, including the question section for 'www.mit.edu'. The packet bytes on the right show the raw data of the DNS query.

No.	Time	Source	Destination	Protocol	Length	Info
15	4.951232	128.238.38.160	128.238.29.22	DNS	86	Standard query 0x0001 PTR 22.29.238.128.in-addr.arpa
16	4.951638	128.238.29.22	128.238.38.160	DNS	118	Standard query response 0x0001 PTR 22.29.238.128.in-addr.arpa PTR dns-prime.poly.edu
17	4.952571	128.238.38.160	128.238.29.22	DNS	80	Standard query 0x0002 A www.mit.edu.poly.edu
18	4.952953	128.238.29.22	128.238.38.160	DNS	139	Standard query response 0x0002 No such name A www.mit.edu.poly.edu SOA dns-prime.poly.edu
19	4.953172	128.238.38.160	128.238.29.22	DNS	71	Standard query 0x0003 A www.mit.edu
20	4.969929	128.238.29.22	128.238.38.160	DNS	196	Standard query response 0x0003 A www.mit.edu A 18.7.22.83 NS BITSY.mit.edu NS STRAMB.mit.edu NS W20NS.mit.edu A 18.72.0.3 A 18.71.0.1

Frame 19: 71 bytes on wire (568 bits), 71 bytes captured (568 bits) on Ethernet II, Src: IBM10:60:99 (00:09:6b:10:60:99), Dst: All-HSRP-routers_00 (00:00:0c:07:ac:00)

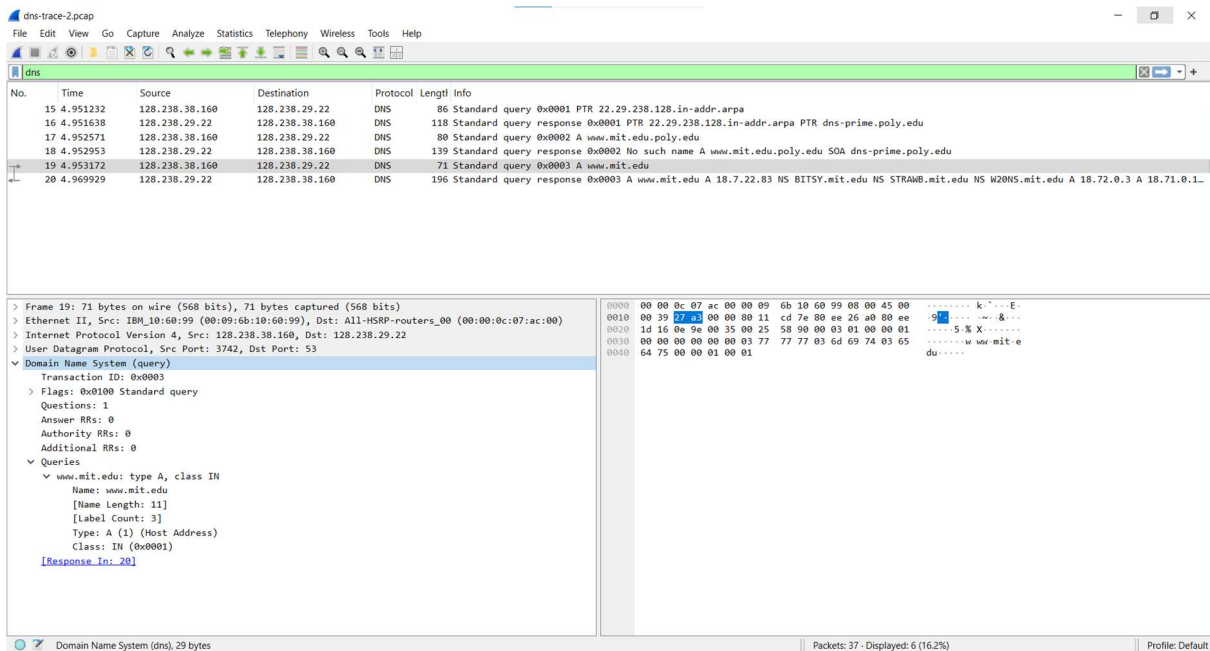
Internet Protocol Version 4, Src: 128.238.38.160, Dst: 128.238.29.22

0100 = Version: 4
.... 0101 = Header Length: 20 bytes (5)
> Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 57
Identification: 0x27a3 (10147)
> 000. = Flags: 0x0
... 0 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: UDP (17)
Header Checksum: 0xcd7e [validation disabled]
[Header checksum status: Unverified]
Source Address: 128.238.38.160
Destination Address: 128.238.29.22
[Stream index: 1]
> User Datagram Protocol, Src Port: 3742, Dst Port: 53
> Domain Name System (query)

0000 00 00 0c 07 ac 00 00 09 6b 10 60 99 00 00 45 00 k...E
0010 00 39 27 a3 00 00 80 11 cd 7e 80 ee 26 a0 80 ee 9'.....&...
0020 1d 16 0e 9e 00 35 00 25 58 00 00 03 01 00 00 01 S % X
0030 00 00 00 00 00 00 03 77 77 03 6d 69 74 03 65 www.mit-e
0040 64 75 00 00 01 00 01 du.....

10. Examine the DNS query message. What “Type” of DNS query is it? Does the query message contain any “answers”? Add screenshots in your answer.

Type “A”.



11. Examine the DNS response message. How many “answers” are provided? What do each of these answers contain? Add screenshots in your answer.

1 answer.

